

Dealer Management System (DMS)

Software Requirement Document (SRD)

1. Project Overview

The Dealer Management System (DMS) is a centralized web and mobile-based platform designed to manage product installations, service requests, spare part replacements, LED replacements, warranty tracking, and dealer operations across India.

The system connects the **Company, Distributors, Dealers/Retailers, Service Engineers**, and **Customers** on a single platform to provide a seamless after-sales service experience.

2. Objective

The objective of the Dealer Management System is to:

- Digitize the complete after-sales service workflow.
 - Manage installation requests.
 - Handle spare part replacement requests.
 - Process product replacement requests.
 - Track warranty services.
 - Auto-assign service requests based on customer PIN Code.
 - Manage dealer payouts.
 - Maintain complete service history.
 - Reduce manual work and improve customer satisfaction.
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3. User Roles

1. Super Admin (Company)

Responsibilities

- Manage Dealers
- Manage Distributors
- Manage Retailers

- Manage Service Engineers
 - Manage Products
 - Manage Customers
 - Approve Installations
 - Approve Warranty Claims
 - Manage Replacement Requests
 - Upload Happy Code
 - Close Service Calls
 - Generate Reports
 - Configure PIN Code Mapping
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2. Distributor

- Manage Dealers
 - View Service Requests
 - View Installations
 - Track Product Replacement
 - View Reports
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3. Dealer / Retailer

- Register Customer
 - Sell Product
 - Request Installation
 - Raise Service Request
 - Upload Bills
 - Request Spare Parts
 - Request product Replacement
 - Track Request Status
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4. Service Engineer

- Accept Assigned Jobs
 - Visit Customer
 - Complete Installation
 - Update Service Report
 - Upload Images
 - Upload Customer Signature
 - Update Product Status
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5. Customer

- Product Registration
 - View Warranty
 - Track Installation
 - Track Service Status
 - Track Replacement Status
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4. Core Services

The system provides three primary services.

Service 1

Product Installation

Process Flow

Retailer



Customer Registration



Product Details Added



Installation Request Submitted



Company Approval



Service Engineer Assigned



Installation Completed



Installation Report Uploaded



Company Verification



Dealer / Engineer Payout



Installation Closed

Data Captured

Customer Name

Mobile Number

Address

PIN Code

State

District

Product

Model Number

Serial Number

Purchase Date

Invoice Upload

Installation Date

Installation Photos

Customer Signature

Engineer Remarks

Approval Status

Payout Status

Service 2

Spare Part Change / Repair Service

This module manages warranty repair and spare part replacement.

Process Flow

Customer Complaint

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Retailer Creates Service Request

↓

Company Reviews Complaint

↓

Company Approves Request

↓

Part Dispatch

OR

Local Purchase

↓

If Local Purchase

Dealer Uploads Bill

↓

Company Verifies Bill

↓

Company Updates Dispatch Details



Happy Code Generated



Customer Confirms Happy Code



Call Closed

Information Stored

Complaint Details

Product Details

Fault Description

Part Required

Part Number

Dispatch Details

Courier Name

Tracking Number

Dispatch Date

Expected Delivery

Local Purchase Bill

Bill Approval

Part Installation Date

Engineer Remarks

Happy Code

Call Closing Date

Customer Feedback

Service 3

Product Replacement

This module handles complete product replacement.

Replacement can be provided by:

- Company
- Dealer / Retailer

Process Flow

Customer Complaint

↓

Dealer Creates Replacement Request

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Company Verification

↓

Replacement Approval

↓

Replacement Product Dispatch

OR

Dealer Replacement

↓

Old Product Return

↓

Replacement Confirmation

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Warranty Updated



Request Closed

Information Captured

Old Product Serial Number

Replacement Product Serial Number

Dispatch Details

Courier Tracking

Replacement Date

Return Status

Warranty Status

Customer Confirmation

Engineer Remarks

Call Closed

5. PAN India Service Management

The application supports operations across India.

Auto Assignment

When a customer registers:

Customer Address



PIN Code



System Finds Assigned Dealer



Service Engineer Assigned Automatically



Installation / Service Created

If no dealer exists for the PIN Code:

- Transfer request to nearest dealer
- Transfer to company support
- Manual assignment available

6. PIN Code Mapping

Each Dealer or Service Engineer can be assigned one or multiple PIN Codes.

Example

Dealer A

302001

302002

302003

Dealer B

305001

305002

305003

When a customer enters PIN Code **302001**, the system automatically assigns the request to Dealer A.

7. Product Return Management

The system supports:

- Return Request
 - Replacement Request
 - Warranty Return
 - Product Pickup
 - Return Tracking
 - Warehouse Receipt
 - Return Approval
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8. Warranty Management

Warranty includes:

- Warranty Start Date
 - Warranty Expiry Date
 - Installation Date
 - Product Registration Date
 - Warranty Status
 - Replacement Warranty
 - Service History
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9. Happy Code Verification

After repair or installation:

Company Generates Happy Code

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Customer Receives Happy Code

↓

Engineer Enters Happy Code

↓

System Verifies Code

↓

Call Closed

This ensures service completion is confirmed by the customer.

10. Payout Management

After successful installation:

Installation Verified



Company Approval



Dealer / Engineer Payout Generated



Payment Status Updated

The system stores:

- Installation Incentive
- Service Charges
- Replacement Charges
- Payment Date
- Payment Status

11. Notifications

Notifications can be sent through:

- SMS
- WhatsApp
- Mobile App Notifications

Events include:

- Installation Assigned
- Service Assigned
- Replacement Approved
- Product Dispatched
- Happy Code Generated
- Call Closed
- Payout Released

12. Reports & Dashboard

Reports available include:

- Installation Report
 - Service Report
 - Replacement Report
 - Dealer Performance
 - Engineer Performance
 - Product-wise Report
 - State-wise Report
 - PIN Code-wise Report
 - Payout Report
 - Warranty Report
 - Customer Report
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13. Document Upload

The system supports uploading:

- Customer Invoice
 - Installation Photos
 - Product Photos
 - Warranty Card
 - Dealer Bill
 - Local Purchase Bill
 - Service Report
 - Customer Signature
 - Replacement Documents
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14. Workflow Summary

Installation

Customer Registration → Installation Request → Company Approval → Engineer Assignment
→ Installation → Verification → Payout → Closed

Spare Part Service

Complaint → Company Approval → Part Dispatch / Local Purchase → Bill Upload → Happy Code → Call Closed

LED Replacement

Complaint → Approval → Replacement Dispatch → Product Return → Warranty Update → Call Closed

15. Future Enhancements

- Mobile Application (Android & iOS)
 - GPS Tracking for Service Engineers
 - Barcode / QR Code Scanning
 - Offline Service Mode
 - AI-based Complaint Categorization
 - Predictive Maintenance
 - Inventory Management
 - Courier API Integration
 - WhatsApp API Integration
 - Dealer Performance Analytics
 - Service SLA Monitoring
 - E-signature Support
 - Multi-language Support
 - Power BI Dashboard Integration
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Conclusion

The Dealer Management System (DMS) provides a complete end-to-end solution for managing installations, spare part services, LED replacements, warranty claims, dealer operations, service engineer assignments, payouts, and product returns across PAN India. With automated PIN Code-based assignment, workflow approvals, document management, and Happy Code verification, the platform ensures efficient after-sales service, improved operational transparency, and enhanced customer satisfaction.